

yolov8m Hyperparameter Tuning Report

Generated: 2025-11-25 01:32:41

1. Optimization Overview

Property	Value
Model	yolov8m
Dataset	bdd100k_yolo_limited
Total Trials	6
Completed Trials	6
Failed Trials	0
Best Trial	4
Best mAP@0.5	0.2794
Optimization Duration	0:05:59.584920

2. Optimization Configuration

Parameter	Value
Total Trials	5
Epochs per Trial	3
Batch Size	64
Image Size	640
Startup Trials (TPE)	2
Device	cuda
Number of Classes	10
Train Images	945
Val Images	611

3. Best Hyperparameters

Parameter	Value	Description
optimizer	SGD	Optimization algorithm
lr0	0.000635	Initial learning rate
momentum	0.939091	SGD momentum / Adam beta1
weight_decay	0.000362	Weight decay (L2 penalty)
warmup_epochs	0	Warmup epochs
warmup_momentum	0.571493	Warmup momentum
warmup_bias_lr	0.028262	
mosaic	0.960024	Mosaic augmentation
mixup	0.040334	Mixup augmentation

4. Top 20 Trials Performance

#	mAP@0.5	Optimizer	lr0	momentum	mixup	mosaic
1	0.2794	N/A	N/A	N/A	N/A	N/A
2	0.0866	N/A	N/A	N/A	N/A	N/A
3	0.0864	N/A	N/A	N/A	N/A	N/A
4	0.0844	N/A	N/A	N/A	N/A	N/A
5	0.0047	N/A	N/A	N/A	N/A	N/A
6	0.0005	N/A	N/A	N/A	N/A	N/A

4.1 Detailed Hyperparameters - Top 5 Trials

Rank 1: Trial 4 (mAP@0.5: 0.2794)

Rank 2: Trial 3 (mAP@0.5: 0.0866)

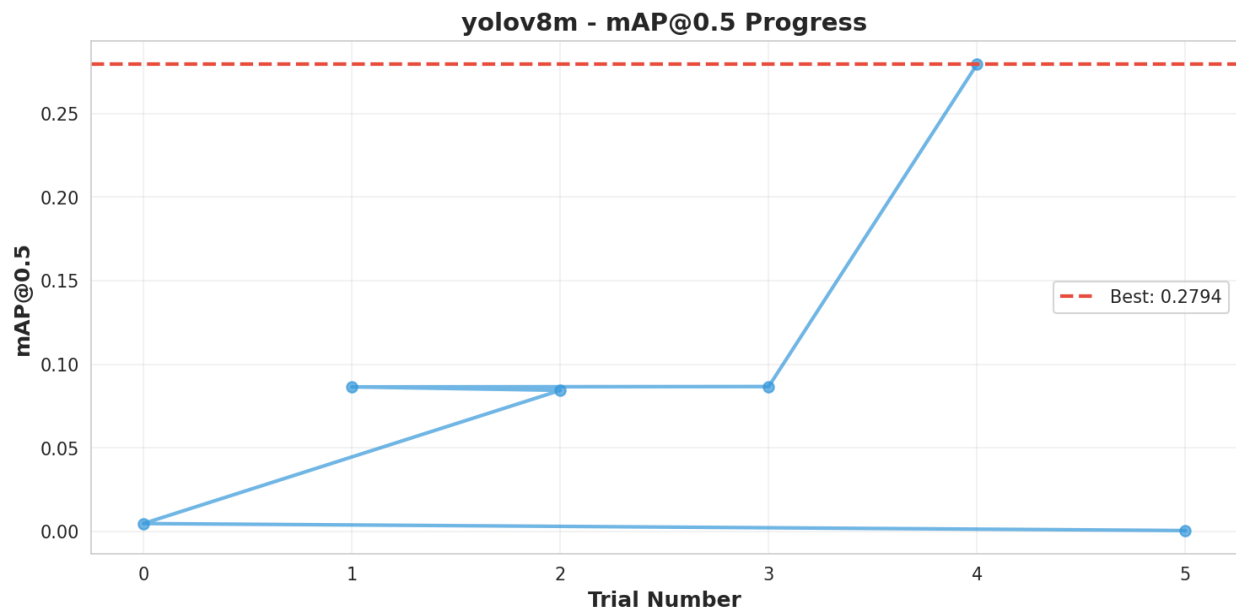
Rank 3: Trial 1 (mAP@0.5: 0.0864)

Rank 4: Trial 2 (mAP@0.5: 0.0844)

Rank 5: Trial 0 (mAP@0.5: 0.0047)

5. Optimization Visualizations

5.1 mAP@0.5 Progress Over Trials



5.2 Learning Rate Impact on Performance

5.3 Optimizer Performance Comparison

5.4 Augmentation Parameters Impact

5.5 Regularization Parameters Impact

6. All Trials Summary

Metric	Value
Completed Trials	6
Best mAP@0.5	0.2794
Worst mAP@0.5	0.0005
Mean mAP@0.5	0.0903
Std Dev mAP@0.5	0.1012
Median mAP@0.5	0.0854